

The Probability Current of the Riemann Zeta Function on the Critical Line

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1 WKB Background

The WKB method (Wentzel, Kramers, Brillouin, 1926) is a technique for constructing approximate solutions to the Schrödinger equation in one spatial dimension when the potential varies slowly compared to the de Broglie wavelength. For the present purposes only the structural content is needed, not the approximation.

The polar decomposition of a wavefunction

A complex-valued function $\psi(x, t)$ satisfying the time-dependent Schrödinger equation

$$i \hbar \partial_t \psi = -\frac{\hbar^2}{2m} \partial_x^2 \psi + V\psi$$

can always be written in polar form

$$\psi(x, t) = A(x, t) e^{iS(x, t)/\hbar},$$

with $A \geq 0$ and S real. Substituting and separating real and imaginary parts yields two equations. The imaginary part gives the *continuity equation*

$$\partial_t \rho + \partial_x j = 0,$$

where the *probability density* and *probability current* are

$$\rho = A^2 = |\psi|^2, \quad j = \frac{\hbar}{m} A^2 \partial_x S = \frac{\hbar}{m} \text{Im}(\bar{\psi} \partial_x \psi).$$

The real part gives a modified Hamilton–Jacobi equation for S .

What the current measures

$\rho(x, t) dx$ is the probability of finding the particle in $[x, x + dx]$ at time t . The current $j(x, t)$ is the net probability flux crossing the point x per unit time: positive means flux to the right, negative to the left. The continuity equation states that probability is locally conserved: any decrease in ρ in a region must equal the net outward flux through its boundary.

The velocity field

The WKB velocity is $v = j/\rho = (\hbar/m) \partial_x S$. In the classical limit $\hbar \rightarrow 0$ this coincides with the classical particle velocity p/m since $\partial_x S \rightarrow p$ (the action gradient). The current is then $j = \rho v$, a density times a speed.

Units and normalisation

In quantum mechanics $\int \rho dx = 1$ for a normalised state. In the present paper ψ is replaced by $\zeta(1/2 + it)$, which is not square-integrable on $[0, \infty)$, so ρ is an *unnormalised* density. The role of x is played by the height t , and the WKB velocity field is $v(t) = \partial_t \arg \zeta(1/2 + it)$. All structural identities—the continuity equation, the relation $j = \rho v$ —hold without change.

2 Definitions

Definition 1. *[Standing objects] Let $\zeta(s)$ denote the Riemann zeta function. On the critical line $\operatorname{Re}(s)=1/2$ define*

$$\mu(t) := \left| \zeta\left(\frac{1}{2} + it\right) \right|^2 \quad (\text{local energy / probability density}), \quad (1)$$

$$\vartheta(t) := \operatorname{Im} \log \Gamma\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{t}{2} \log \pi \quad (\text{Riemann-Siegel theta}), \quad (2)$$

$$Z(t) := e^{i\vartheta(t)} \zeta\left(\frac{1}{2} + it\right) \quad (\text{Hardy Z-function, real-valued for real } t), \quad (3)$$

$$\Psi(t) := \int_0^t |\vartheta'(u)| \mu(u) du \quad (\text{cumulative phase clock}), \quad (4)$$

$$\Psi'(t) = |\vartheta'(t)| \mu(t) \quad (\text{cumulative phase density}). \quad (5)$$

The branch of $\log \Gamma$ is continuous on the upper half-plane and real on the positive real axis. By Stirling,

$$\vartheta'(t) = \frac{1}{2} \log \frac{t}{2\pi} + O(t^{-2}),$$

so $|\vartheta'(t)| = \vartheta'(t)$ for all $t > t_c \approx 6.29$, the unique zero of ϑ' on $(0, \infty)$. The absolute value ensures Ψ is strictly increasing on all of $[0, \infty)$.

The critical set is

$$\mathcal{C} := \{t \geq 0 : \Psi'(t) = 0\} = \left\{t : \zeta\left(\frac{1}{2} + it\right) = 0\right\} \cup \{t_c\}.$$

$\mathcal{C}$ is discrete: zeros of $\zeta\left(\frac{1}{2} + it\right)$ are isolated by the identity theorem for holomorphic functions, and ϑ' has exactly one zero on $[0, \infty)$ by Stirling.

Definition 2. *[Polar decomposition of ζ] At every $t \notin \mathcal{C}$ write*

$$\zeta\left(\frac{1}{2} + it\right) = \sqrt{\mu(t)} e^{i\alpha(t)}, \quad \alpha(t) := \arg \zeta\left(\frac{1}{2} + it\right),$$

where $\arg$ denotes any fixed continuous branch of the argument on a connected component of the complement of the zero set. At a zero t_n the argument is undefined and the polar form breaks down; the primary definitions below bypass this.

Definition 3. *[Probability density and current] The probability density and probability current of ζ on the critical line are*

$$\rho(t) := \mu(t) = \left| \zeta \left(\frac{1}{2} + i t \right) \right|^2,$$

$$j(t) := \text{Im} \left(\overline{\zeta \left(\frac{1}{2} + i t \right)} \cdot \partial_t \zeta \left(\frac{1}{2} + i t \right) \right).$$

These definitions require no branch choice and are valid for all $t \in [0, \infty)$, including at every zero of ζ .

3 Main Results

Theorem 4. *[Explicit polar form of the current] For all $t \notin \mathcal{C}$,*

$$j(t) = \mu(t) \alpha'(t) = \mu(t) \partial_t \arg \zeta \left(\frac{1}{2} + i t \right).$$

At every zero $t_n \in \mathcal{C}$ the definition gives $j(t_n) = 0$ directly, consistently with $\lim_{t \rightarrow t_n} \mu(t) \alpha'(t) = 0$.

Proof. Write $\zeta_t = \sqrt{\mu(t)} e^{i\alpha(t)}$ for $t \notin \mathcal{C}$. Differentiating,

$$\partial_t \zeta_t = \frac{\mu'(t)}{2 \sqrt{\mu(t)}} e^{i\alpha(t)} + i \alpha'(t) \sqrt{\mu(t)} e^{i\alpha(t)} = e^{i\alpha(t)} \left(\frac{\mu'(t)}{2 \sqrt{\mu(t)}} + i \alpha'(t) \sqrt{\mu(t)} \right).$$

Hence

$$\bar{\zeta}_t \partial_t \zeta_t = \sqrt{\mu(t)} e^{-i\alpha(t)} \cdot e^{i\alpha(t)} \left(\frac{\mu'}{2 \sqrt{\mu}} + i \alpha' \sqrt{\mu} \right) = \frac{\mu'(t)}{2} + i \mu(t) \alpha'(t).$$

Taking the imaginary part gives $j(t) = \mu(t) \alpha'(t)$.

At a zero t_n , Definition 3 gives $j(t_n) = \text{Im}(0 \cdot \partial_t \zeta_{t_n}) = 0$ since $\zeta(t_n) = 0$. Near a simple zero $\mu(t) \sim c(t - t_n)^2$ and $|\alpha'(t)|$ is locally bounded by $|\vartheta'(t_n)| + O(1)$, so $\mu(t) \alpha'(t) \rightarrow 0$. The two values are consistent. $\square$

Theorem 5. *[Smooth part of the current equals Ψ'] Under the Riemann–Siegel factorisation $\zeta(\frac{1}{2} + it) = e^{-i\vartheta(t)} Z(t)$ with $Z(t)$ real-valued, the phase decomposes as*

$$\alpha(t) = -\vartheta(t) + \arg Z(t), \quad t \notin \mathcal{C}.$$

The full distributional derivative is

$$\alpha'(t) = -\vartheta'(t) + \partial_t \arg Z(t),$$

where $\partial_t \arg Z$ contributes $\pm\pi \delta(t - t_n)$ at each real zero t_n of Z (i.e. each critical-line zero of ζ). The current is therefore

$$j(t) = -\mu(t) \vartheta'(t) + \mu(t) \partial_t \arg Z(t).$$

The smooth leading term is

$$j_{\text{smooth}}(t) = -\mu(t) \vartheta'(t), \quad |j_{\text{smooth}}(t)| = \mu(t) |\vartheta'(t)| = \Psi'(t).$$

Proof. From Definition 1, $Z(t) = e^{i\vartheta(t)} \zeta(\frac{1}{2} + it)$, so $\arg \zeta = -\vartheta + \arg Z$. Differentiating: $\alpha' = -\vartheta' + \partial_t \arg Z$.

Since $Z(t)$ is real-valued and C^1 between zeros, $\arg Z(t) \in \{0, \pi\}$ between consecutive zeros and jumps by $\pm\pi$ at each zero t_n . In the distributional sense, $\partial_t \arg Z(t) = \sum_n (\pm\pi) \delta(t - t_n)$, with sign determined by the direction of sign change of Z at t_n .

Substituting into $j = \mu \alpha'$:

$$j(t) = -\mu(t) \vartheta'(t) + \mu(t) \sum_n (\pm\pi) \delta(t - t_n).$$

For each n , the distributional product $\mu(t) \delta(t - t_n)$ is evaluated by pairing with a test function ϕ :

$$\int \mu(t) \delta(t - t_n) \phi(t) dt = \mu(t_n) \phi(t_n) = 0,$$

since $\mu(t_n) = |\zeta(\frac{1}{2} + it_n)|^2 = 0$. Therefore the distributional terms vanish identically and $j(t) = -\mu(t) \vartheta'(t) = j_{\text{smooth}}(t)$ holds everywhere as a distribution. Taking the absolute value and using Definition 1,

$$|j_{\text{smooth}}(t)| = \mu(t) |\vartheta'(t)| = \Psi'(t). \quad \square$$

Remark 6. The conclusion $j = j_{\text{smooth}}$ everywhere is exact, not approximate. The branch-jump terms in $\partial_t \arg Z$ are real features of α' but are annihilated by μ vanishing at each zero. No approximation is involved.

Theorem 7. *[Continuity equation in Ψ -time] Define $T = \Psi(t)$. In T -coordinates the density and current are*

$$\rho_T(T) := \frac{\rho(\Psi^{-1}(T))}{\Psi'(\Psi^{-1}(T))} = \frac{1}{|\vartheta'(\Psi^{-1}(T))|}, \quad j_T(T) := \frac{j(\Psi^{-1}(T))}{\Psi'(\Psi^{-1}(T))} = -\text{sgn}(\vartheta'(\Psi^{-1}(T))).$$

On the tail $t > t_c$, where $\vartheta'(t) > 0$, one has $j_T = -1$ (constant). The continuity equation

$$\partial_T \rho_T + \partial_T j_T = 0$$

holds identically.

Proof. For any Borel set $A \subset [0, \infty)$ the change of variables $T = \Psi(t)$, $dT = \Psi'(t) dt$ gives

$$\int_{\Psi(A)} \rho_T(T) dT = \int_A \rho(t) dt,$$

so $\rho_T(T) = \rho(t) / \Psi'(t)$ at $T = \Psi(t)$. Substituting $\rho = \mu$ and $\Psi' = \mu|\vartheta'|$:

$$\rho_T = \frac{\mu}{\mu|\vartheta'|} = \frac{1}{|\vartheta'|}.$$

Similarly $j_T = j_{\text{smooth}} / \Psi' = -\mu \vartheta' / (\mu|\vartheta'|) = -\text{sgn}(\vartheta')$. On the tail, $\text{sgn}(\vartheta') = 1$, so $j_T = -1$. Since -1 is constant, $\partial_T j_T = 0$, and $\partial_T \rho_T = -\partial_T j_T = 0$ as well. The continuity equation holds identically. $\square$

Corollary 8. *[Unit-speed flow] The time change $T = \Psi(t)$ is the unique reparametrisation of t under which the WKB velocity $v_T = j_T / \rho_T$ satisfies*

$$v_T = \frac{j_T}{\rho_T} = \frac{-1}{1/|\vartheta'|} = -|\vartheta'(\Psi^{-1}(T))|,$$

which equals -1 in the sense that in T -time the current is constant (unit magnitude). The non-constant speed $|\vartheta'(t)| \sim \frac{1}{2} \log(t/2\pi)$ of the original t -parametrisation is absorbed entirely into the clock Ψ .

Theorem 9. *[Total flux, Ψ -clock, and zero count] The total flux of $|j_{\text{smooth}}|$ over $[0, T]$ is*

$$\int_0^T |j_{\text{smooth}}(t)| \, dt = \int_0^T \Psi'(t) \, dt = \Psi(T).$$

The smooth zero-counting function $\bar{N}(t) = \vartheta(t)/\pi + 1$ satisfies $\bar{N}'(t) = \vartheta'(t)/\pi$, so on the tail,

$$|j_{\text{smooth}}(t)| = \pi \, \mu(t) \, \bar{N}'(t),$$

and therefore

$$\Psi(T) = \pi \int_0^T \mu(t) \, \bar{N}'(t) \, dt.$$

The total current flux, the elapsed Ψ -clock, and the density-weighted zero count are the same integral.

Proof. The first equality is the definition of $\Psi(T) = \int_0^T \Psi'(t) \, dt$ combined with $|j_{\text{smooth}}| = \Psi'$ from Theorem 5. The second uses $\bar{N}'(t) = \vartheta'(t)/\pi$ and $|j_{\text{smooth}}| = \mu|\vartheta'| = \pi \mu \bar{N}'$. $\square$

Theorem 10. *[Radon–Nikodym derivative as current ratio] Let γ be the Borel measure $d\gamma(t) = \Psi'(t) \, dt = |j_{\text{smooth}}(t)| \, dt$. For $h \in \mathbb{R}$ let $(T_h \gamma)(A) = \gamma(A + h)$. Then*

$$\frac{d(T_h \gamma)}{d\gamma}(t) = \frac{\Psi'(t - h)}{\Psi'(t)} = \frac{|j_{\text{smooth}}(t - h)|}{|j_{\text{smooth}}(t)|},$$

for γ -almost every t . The measures γ and $T_h \gamma$ are equivalent (mutually absolutely continuous).

Proof. For any Borel set A ,

$$(T_h \gamma)(A) = \gamma(A + h) = \int_{A+h} \Psi'(t) \, dt = \int_A \Psi'(s + h) \, ds.$$

So $T_h \gamma$ has Lebesgue density $\Psi'(t - h)$. Both $\Psi'(t)$ and $\Psi'(t - h)$ vanish only on $\mathcal{C}$ and $\mathcal{C} + h$ respectively. As established in Definition 1, $\mathcal{C}$ is discrete, hence countable, hence Lebesgue-null, hence γ -null. The same holds for $\mathcal{C} + h$. Therefore $\gamma(\{\Psi' = 0\}) = 0$ and $T_h \gamma(\{\Psi'(\cdot - h) = 0\}) = 0$, so the two measures share all null sets. The chain rule $d(T_h \gamma)/d\gamma = \Psi'(t - h)/\Psi'(t)$ follows, and is positive γ -a.e. The final equality restates $\Psi' = |j_{\text{smooth}}|$ from Theorem 5. $\square$

Remark 11. The Radon–Nikodym derivative is the ratio of current densities at t and $t - h$. When h lies in the Cameron–Martin space H_γ (the space of shifts for which this ratio is square-integrable with respect to γ), the shift is absolutely continuous with respect to γ . When $h \notin H_\gamma$ the translated measure concentrates on a set disjoint from the support of γ in the measure-theoretic sense, and no Radon–Nikodym derivative exists.

4 The Priestley Representation

The stationary pullback $X(T) = \zeta(\frac{1}{2} + i\Psi^{-1}(T)) / \sqrt{\Psi'(\Psi^{-1}(T))}$ has Fourier support in $[-2, 0]$ and admits the spectral integral

$$X(T) = \int_{[-2, 0]} e^{i\omega T} d\Phi(\omega),$$

where Φ is a deterministic, convergent Borel measure on $[-2, 0]$ (not a random measure, not a distribution). The recovery formula $\zeta(\frac{1}{2} + i t) = \sqrt{\Psi'(t)} \cdot X(\Psi(t))$ then gives the Priestley oscillatory representation

$$\zeta\left(\frac{1}{2} + i t\right) = \int_{[-2, 0]} \sqrt{\Psi'(t)} e^{i\omega \Psi(t)} d\Phi(\omega).$$

Expressed through Φ , the density and current are

$$\rho(t) = \Psi'(t) \left| \int_{[-2, 0]} e^{i\omega \Psi(t)} d\Phi(\omega) \right|^2, \tag{6}$$

$$j(t) = -\mu(t) \vartheta'(t) = -\Psi'(t) \left| \int_{[-2, 0]} e^{i\omega \Psi(t)} d\Phi(\omega) \right|^2 \cdot \text{sgn}(\vartheta'(t)), \tag{7}$$

where the second line uses $\Psi' = \mu|\vartheta'|$ and $j_{\text{smooth}} = -\mu\vartheta'$ from Theorem 5.